MR-1: Literature Extraction & Convention Fixing for Lorentzian Continuation of Spectral Causal Theory

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Abstract

This document collects all definitions, theorems, and results from the literature needed to establish the Lorentzian continuation of Spectral Causal Theory (SCT). The central obstacle is that the Euclidean spectral action $\mathrm{Tr}\, f(D^2/\Lambda^2)$ relies on D^2 being a positive elliptic operator, which fails in Lorentzian signature where D^2 is hyperbolic. We survey seven topic areas: (1) Krein-space formalism and Lorentzian spectral triples; (2) Lorentzian spectral geometry and index theorems; (3) infinite-derivative gravity and ghost-freedom criteria; (4) Lee-Wick mechanism and fakeon quantization; (5) Wick rotation in curved spacetime; (6) the Chamseddine-Connes spectral action; (7) causal-set d'Alembertians. For each topic, we extract the precise mathematical definitions and results, identify which assumptions are satisfied by SCT and which constitute gaps, and assess the applicability to the MR-1 problem. All citations carry real arXiv identifiers.

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1 Introduction and Scope

1.1 The Euclidean–Lorentzian Gap

The SCT spectral action is defined in Euclidean signature via

$$S_{\text{spec}} = \mathrm{Tr}\Big(f(D^2/\Lambda^2)\Big)\,, \tag{1}$$

where D is the (Euclidean) Dirac operator and f is a suitable cutoff function. In the heat-kernel expansion at one loop and to order $\mathcal{R}^2$, this produces the effective gravitational action with nonlocal form factors $F_1(z)$, $F_2(z)$ that have been computed and verified in NT-1 through NT-4a.

In Lorentzian signature $(-,+,+,+)$, the operator D^2 is no longer elliptic or positive: the principal symbol has indefinite sign, the heat kernel e^{-tD^2} is not trace-class, and (1) is undefined. The task of MR-1 is to find a mathematically rigorous Lorentzian formulation that reduces to the known Euclidean results in the appropriate limit.

1.2 SCT Context: Verified Euclidean Results

From NT-1 through NT-4a, the following results are established and verified:

- Master function: $\phi(z) = \int_0^1 e^{-\xi(1-\xi)z}\,\mathrm{d}\xi$, entire of order 1, type 1/4.
- Spin-2 propagator denominator: $\Pi_{\mathrm{TT}}(z) = 1 + \frac{13}{60}\,z\,\hat{F}_1(z)$.
- Spin-0 propagator denominator: $\Pi_s(z;\xi) = 1 + 6(\xi - \frac{1}{6})^2\,z\,\hat{F}_2(z;\xi)$.
- First positive-real zero of Π_{TT}: $z_0 \approx 2.41484$ with residue $R_2 \approx -0.493$.
- Local limit coefficients: $\alpha_C = 13/120$ (Weyl^2), $\alpha_R(\xi) = 2(\xi - \frac{1}{6})^2$ (Ricci scalar^2).
- Scalar mode decouples at $\xi = 1/6$ (conformal coupling).

1.3 Document Conventions

We follow the conventions established in the NT-1 and NT-2 literature documents:

- Signature: $(-,+,+,+)$ for Lorentzian, $(+,+,+,+)$ for Euclidean.
- Riemann tensor: $R^\rho{}_{\sigma\mu\nu} = \partial_\mu \Gamma^\rho{}_{\nu\sigma} - \cdots$ (MTW sign).
- Dirac operator: $D = -\mathrm{i}\gamma^\mu \nabla_\mu$ in Lorentzian, $D_E = \gamma_E^\mu \nabla_\mu$ in Euclidean.
- Spectral parameter: $z = \Box/\Lambda^2$ (Euclidean), $z = -\Box_L/\Lambda^2$ (Lorentzian Wick-rotated).
- Annotations: [paper ref] for literature sources, [GAP] for identified gaps, [our convention] for our conventions.

2 Krein Spaces and Lorentzian Spectral Triples

2.1 Krein Space Fundamentals

Definition 2.1 (Krein space)**.** [Bognár 1974; Strohmaier, math-ph/0110001] A **Krein space** is a pair $(\mathcal{K}, [\cdot,\cdot])$ where $\mathcal{K}$ is a complex topological vector space and $[\cdot,\cdot] : \mathcal{K}\times\mathcal{K} \to \mathbb{C}$ is a non-degenerate sesquilinear form (not necessarily positive-definite), such that there exists a **fundamental symmetry** J satisfying:

(i) $J^2 = \mathrm{Id}, \quad J = J^*$ (with respect to $[\cdot,\cdot]$),

(ii) $\langle\psi,\phi\rangle := [\psi, J\phi]$ is a positive-definite inner product making $\mathcal{K}$ into a Hilbert space.

The eigenspaces $\mathcal{K}_\pm = \ker(J \mp \mathrm{Id})$ give the **fundamental decomposition** $\mathcal{K} = \mathcal{K}_+ \oplus \mathcal{K}_-$, where $[\cdot,\cdot]$ is positive-definite on $\mathcal{K}_+$ and negative-definite on $\mathcal{K}_-$.

Definition 2.2 (J-self-adjointness)**.** [Strohmaier, math-ph/0110001, Def. 2.1] An operator D on a Krein space $(\mathcal{K}, [\cdot,\cdot])$ is **Krein-self-adjoint** (or **J-self-adjoint**) if

$$[D\psi, \phi] = [\psi, D\phi] \quad \forall\, \psi, \phi \in \mathrm{Dom}(D)\,. \tag{2}$$

Equivalently, $JDJ = D^*$, where D^* is the Hilbert-space adjoint with respect to $\langle\cdot,\cdot\rangle = [\cdot, J\cdot]$.

Remark 2.3 (Physical interpretation). [Strohmaier, math-ph/0110001, §3] For the Lorentzian Dirac operator on (M, g) with g of signature $(-,+,+,+)$:

- The Krein inner product is the Dirac inner product: $[\psi, \phi] = \int_M \bar{\psi}\phi \,\mathrm{dvol}_g$.
- The fundamental symmetry is $J = \mathrm{i}\gamma^0$ (or γ^0 depending on conventions for γ^0 Hermiticity).
- The Hilbert-space inner product is $\langle\psi, \phi\rangle = \int_M \psi^\dagger \phi \,\mathrm{dvol}_g$ (positive-definite).
- $D = -\mathrm{i}\gamma^\mu \nabla_\mu$ is J-self-adjoint but NOT Hilbert-space self-adjoint.

2.2 Pseudo-Riemannian Spectral Triples

Definition 2.4 (Pseudo-Riemannian spectral triple)**.** [Strohmaier, math-ph/0110001, Def. 3.1; Paschke-Sitarz, math-ph/0611029] A **pseudo-Riemannian spectral triple** is a tuple $(\mathcal{A}, \mathcal{K}, D, J)$ where:

(i) $\mathcal{A}$ is a unital $*$-algebra represented faithfully on $\mathcal{K}$,
(ii) $(\mathcal{K}, [\cdot,\cdot])$ is a Krein space with fundamental symmetry J,
(iii) D is a densely defined J-self-adjoint operator on $\mathcal{K}$,
(iv) $[D, a]$ is bounded for all $a \in \mathcal{A}$.

This replaces the Connes axioms where $\mathcal{H}$ is a Hilbert space and D is self-adjoint.

Remark 2.5 (Comparison with Riemannian spectral triple). The Riemannian spectral triple $(\mathcal{A}, \mathcal{H}, D)$ with $D = D^*$ on a Hilbert space $\mathcal{H}$ is a special case where $J = \mathrm{Id}$ (trivial fundamental symmetry), so $\mathcal{K} = \mathcal{H}$, $[\cdot,\cdot] = \langle\cdot,\cdot\rangle$, and Krein-self-adjointness reduces to ordinary self-adjointness.

The key structural difference in the Lorentzian case is that D^2 has *indefinite* spectrum: it acts as a hyperbolic operator $(-\partial_t^2 + \Delta + \text{curvature})$, with eigenvalues unbounded in both directions. In particular, the heat semigroup e^{-tD^2} is *not* trace-class.

2.3 Van den Dungen–Rennie: Lorentzian NCG with Wick Rotation

Definition 2.6 (Lorentzian spectral triple with Wick rotation)**.** [van den Dungen & Rennie, arXiv:1503.06916, Def. 3.2] A **Lorentzian spectral triple** is a pseudo-Riemannian spectral triple $(\mathcal{A}, \mathcal{K}, D, J)$ equipped with a **Wick rotation** operator β satisfying:

(i) $\beta = \beta^*$ and $\beta > 0$ (positive, self-adjoint on $\mathcal{H} = (\mathcal{K}, \langle\cdot,\cdot\rangle)$),
(ii) $D_E := \beta D \beta^{-1}$ is self-adjoint on $\mathcal{H}$ (the **Euclidean Dirac operator**),
(iii) Compatibility with the real structure: β commutes with J_F (finite part) and anti-commutes with γ_F (grading), or vice versa depending on KO-dimension.

Remark 2.7 (Explicit β for manifolds). [van den Dungen & Rennie, arXiv:1503.06916, §4] For a globally hyperbolic Lorentzian manifold (M, g) with a global time function t, the Wick rotation operator is $\beta = (N/\sqrt{|g_{00}|})^{1/2}$ acting on spinors, where N is the lapse function. In the static case with $g_{00} = -1$: $\beta = 1$ and $D_E = D_L$ up to sign — the Euclidean and Lorentzian operators coincide.

For general spacetimes, β depends on the choice of foliation, and the definition is coordinate-dependent. This is a known limitation.

2.4 Barrett: KO-Dimension and Fermion Doubling

Theorem 2.8 (Barrett's Lorentzian KO-dimension). *[Barrett, J. Phys. A **40** (2007) 9911, arXiv:hep-th/0608221, Thm. 4.1] The finite spectral triple for the Standard Model admits a real structure of **KO-dimension** 2 (mod 8) (equivalently 6 (mod 8) for Lorentzian signature), such that:*

(i) The fermionic action $S_f = [J\Psi, D_A\Psi]$ (Krein inner product) reproduces the full Standard Model fermionic Lagrangian,

(ii) No fermion doubling is required: the fermion space is $\mathcal{H}_F = \mathbb{C}^{96}$ (one generation: 16 states × 3 colours × 2 chiralities), not doubled,

(iii) The grading γ_F and real structure J_F satisfy the KO-dimension 6 sign table:

$$J_F^2 = \mathrm{Id}, \quad J_F D_F = D_F J_F, \quad J_F \gamma_F = -\gamma_F J_F\,. \tag{3}$$

Remark 2.9 (Contrast with Euclidean KO = 0). In the Euclidean formulation (KO-dimension 0 (mod 8)), the sign table requires $J_F\gamma_F = +\gamma_F J_F$, which forces fermion doubling to accommodate the real structure. The Lorentzian KO = 6 flips this sign, eliminating the doubling.

Barrett's result demonstrates that the fermionic sector of the spectral action has a natural Lorentzian formulation. The bosonic sector (spectral action $\mathrm{Tr}\, f(D^2/\Lambda^2)$) remains the open problem.

2.5 Franco–Eckstein: Algebraic Causality

Definition 2.10 (Causal cone in Lorentzian NCG). [Franco & Eckstein, Symmetry **6** (2014) 1, arXiv:1212.5171, Def. 3.1] In a Lorentzian spectral triple $(\mathcal{A}, \mathcal{K}, D, J)$, the **causal cone** $\mathcal{C}$ is the set of Hermitian elements $a \in \mathcal{A}_{\mathrm{sa}}$ such that $\mathrm{i}[D, a]$ is a non-negative operator on $\mathcal{K}$ (in the Krein sense):

$$\mathcal{C} = \{a \in \mathcal{A}_{\mathrm{sa}} : [\psi, \mathrm{i}[D, a]\psi] \geq 0 \;\; \forall\, \psi \in \mathcal{K}\}\,. \tag{4}$$

The **causal relation** $x \leq y$ between pure states is: $\omega_x(a) \leq \omega_y(a)$ for all $a \in \mathcal{C}$.

Theorem 2.11 (Recovery of commutative causality). *[Franco & Eckstein, arXiv:1212.5171, Thm. 4.2] For the commutative Lorentzian spectral triple $(\mathcal{A} = C^\infty(M), \mathcal{K} = L^2(M, S), D = -\mathrm{i}\gamma^\mu\nabla_\mu)$ on a globally hyperbolic spacetime (M, g), the algebraic causal cone $\mathcal{C}$ recovers the standard Lorentzian causal structure: $x \leq y$ if and only if there exists a future-directed causal curve from x to y.*

3 Lorentzian Spectral Geometry: Index Theorems

Theorem 3.1 (Bär–Strohmaier Lorentzian index theorem). *[Bär & Strohmaier, Amer. J. Math. **141** (2019) 1421, arXiv:1506.00959, Thm. 1.1] Let (M, g) be a compact globally hyperbolic Lorentzian 4-manifold with spin structure and spacelike boundary $\partial M = \Sigma_0 \sqcup \Sigma_1$ (initial and final Cauchy surfaces). With Atiyah-Patodi-Singer (APS) type boundary conditions, the Lorentzian Dirac operator D is a **Fredholm** operator, and*

$$\mathrm{ind}(D) = \int_M \hat{A}(M) - \frac{1}{2}\Big(\eta(\Sigma_0) + \eta(\Sigma_1)\Big)\,, \tag{5}$$

where $\hat{A}(M)$ is the $\hat{A}$-genus density and $\eta(\Sigma_i)$ are the η-invariants of the induced Dirac operators on the boundary Cauchy surfaces.

Remark 3.2 (Equality with Riemannian index). The formula (5) is *identical* to the Atiyah-Patodi-Singer index theorem for the Riemannian case. This is a non-trivial result: it shows that the topological content of the Dirac operator is *signature-independent*.

The key technical ingredient is that the Lorentzian Dirac operator, while not self-adjoint in the Hilbert-space sense, is Fredholm (finite kernel and cokernel) when equipped with APS boundary conditions.

Remark 3.3 (Implications for the spectral action). The index theorem uses only the *kernel* of D (and its formal adjoint), not the full spectrum. The spectral action $\mathrm{Tr}\, f(D^2/\Lambda^2)$ depends on the *entire* spectrum. Therefore, the Bär–Strohmaier theorem establishes that the topological sector is well-defined in Lorentzian signature, but says nothing about the non-topological (curvature-dependent) terms in the spectral action.

[GAP] Extending the spectral action beyond the topological sector to Lorentzian signature is the core open problem of MR-1.

4 Infinite-Derivative Gravity and Ghost-Freedom

4.1 Ghost-Freedom in Nonlocal Gravity

Definition 4.1 (Ghost-free propagator (BMS criterion)). [Biswas, Mazumdar, Siegel, JCAP **0603** (2006) 009, hep-th/0508194, Eq. (7)] Consider the gravitational action with nonlocal form factors:

$$S = \int \mathrm{d}^4x\,\sqrt{g}\left[\frac{R}{16\pi G} + R\,F_R(\Box/\Lambda^2)\,R + C_{\mu\nu\rho\sigma}\,F_C(\Box/\Lambda^2)\,C^{\mu\nu\rho\sigma} + \cdots\right]. \tag{6}$$

The spin-2 propagator around flat space takes the form

$$G_{\mathrm{TT}}^{-1}(k^2) = k^2\Big[1 + c_2\,k^2\,a(k^2/\Lambda^2)\Big]\,, \tag{7}$$

where $a(z)$ depends on F_C. The propagator is **ghost-free** if $1 + c_2\,z\,a(z) \neq 0$ for all $z > 0$ (no additional real poles beyond $k^2 = 0$).

Theorem 4.2 (Tomboulis: exponential entire form factor). *[Tomboulis, hep-th/9702146, §2–3; Modesto, arXiv:1107.2403, Thm. 1]* ***Sufficient condition for ghost-freedom:*** *If $a(z) = e^{H(z)}$ where $H(z)$ is an entire function with $H(0) = 0$, then:*

(i) The propagator $G(k^2) \propto k^{-2}\,e^{-H(k^2/\Lambda^2)}$ has no additional poles (ghost-free),

(ii) The theory is power-counting super-renormalizable for $H(z) \sim z^n$, $n \geq 2$,
(iii) UV behaviour is exponentially suppressed: $G(k^2) \sim k^{-2}\, e^{-|k|^{2n}/\Lambda^{2n}} \to 0$ as $|k| \to \infty$.

Remark 4.3 (BMS-Tomboulis-Modesto programme). The standard choices in the literature are:

$$H(z) = z \quad \text{(Krasnikov, Kuz'min, Tomboulis)}, \tag{8}$$

$$H(z) = \left(e^z - 1\right)/z \quad \text{(Biswas–Mazumdar–Siegel)}, \tag{9}$$

$$H(z) = z^n,\ n \geq 2 \quad \text{(Modesto super-renormalizable)}. \tag{10}$$

All share the property: $a(z) = e^{H(z)}$ has no zeros (the exponential of an entire function never vanishes), hence the propagator denominator $1 + c_2\, z\, a(z)$ cannot vanish for $z > 0$ if $c_2 > 0$ and $a(z) > 0$.

4.2 SCT Form Factors vs. Ghost-Free Condition

Convention 4.4 (SCT propagator denominators)**.** [our convention] [NT-4a, verified] The SCT propagator denominators are:

$$\Pi_{\mathrm{TT}}(z) = 1 + \frac{13}{60}\, z\, \hat{F}_1(z)\,, \tag{11}$$

$$\Pi_s(z;\xi) = 1 + 6\left(\xi - \frac{1}{6}\right)^2 z\, \hat{F}_2(z;\xi)\,, \tag{12}$$

where $\hat{F}_i(z) = F_i(z)/F_i(0)$ are normalized form factors and $z = \Box_E/\Lambda^2$ is the Euclidean spectral parameter.

Remark 4.5 (SCT does NOT satisfy the BMS ghost-free criterion). [GAP] The SCT form factors are *derived* from the heat kernel, not *engineered* to be ghost-free. Specifically:

1. $F_1(z)$ and $F_2(z)$ are expressed through the master function $\phi(z) = \int_0^1 e^{-\xi(1-\xi)z}\, \mathrm{d}\xi$, which is entire of order 1, type 1/4.
2. $\hat{F}_1(z) \neq e^{H(z)}$ for any entire H. The form factors are *not* of exponential type in the sense of Tomboulis's theorem.
3. $\Pi_{\mathrm{TT}}(z_0) = 0$ at $z_0 \approx 2.41484$ (MR-2 d.5, verified to 30 digits).
4. The residue at z_0 is $R_2 = 1/\Pi'_{\mathrm{TT}}(z_0) \approx -0.493$ (negative = ghost).

The SCT ghost pole is *real* and has residue $|R_2| = 0.493$, compared to $|R_2^{\mathrm{Stelle}}| = 1$ for Stelle gravity. This is a 50.7% suppression of the ghost residue.

5 Lee-Wick Mechanism and Fakeon Quantization

5.1 Lee-Wick Propagators

Definition 5.1 (Lee-Wick propagator)**.** [Lee & Wick, Nucl. Phys. B **9** (1969) 209] A **Lee-Wick propagator** is a propagator of the form

$$G(k^2) = \frac{1}{k^2 - m^2 + \mathrm{i}\epsilon} - \frac{1}{k^2 - M^2 + \mathrm{i}\epsilon'}\,, \tag{13}$$

where $M > m$ is the Lee-Wick partner mass. The second pole has *negative residue* (ghost), but the theory maintains perturbative unitarity through the **CLOP prescription**.

Definition 5.2 (CLOP prescription)**.** [Cutkosky, Landshoff, Olive, Polkinghorne, Nucl. Phys. B **12** (1969) 281] The **CLOP prescription** for the ghost pole at $k^2 = M^2$:

(i) In the complex k^0-plane, the ghost pole is displaced by $+\mathrm{i}\epsilon'$ (opposite to the normal Feynman prescription).

(ii) The integration contour is deformed to run between the normal and ghost poles, avoiding both.

(iii) In Cutkosky cutting rules, the ghost contributes with opposite sign, leading to cancellations that preserve unitarity order by order in perturbation theory.

Remark 5.3 (Lee-Wick in higher-derivative gravity). [Grinstein, O'Connell, Wise, arXiv:0710.5528] Stelle's fourth-derivative gravity is equivalent to a Lee-Wick theory: the action $R + \alpha C_{\mu\nu\rho\sigma}C^{\mu\nu\rho\sigma} + \beta R^2$ propagates a massless spin-2 (graviton), a massive spin-2 (ghost, mass $m_2 = \Lambda/\sqrt{2\alpha}$), and a massive spin-0 (mass $m_0 = \Lambda/\sqrt{6\beta}$).

For SCT in the *local limit* (form factors $\to$ constants): $\alpha = 13/120$, $\beta = 2(\xi - 1/6)^2$. Hence:

$$m_2^{\mathrm{Stelle}} = \Lambda\sqrt{60/13} \approx 2.148\,\Lambda\,, \tag{14}$$

$$m_0^{\mathrm{Stelle}}(\xi = 0) = \Lambda\sqrt{6} \approx 2.449\,\Lambda\,. \tag{15}$$

In the full nonlocal theory, the ghost mass is shifted to $m_2^{\mathrm{SCT}} = \Lambda\sqrt{z_0} \approx 1.554\,\Lambda$ (where $z_0 \approx 2.415$).

5.2 Anselmi-Piva Fakeon Prescription

Definition 5.4 (Fakeon quantization)**.** [Anselmi & Piva, JHEP **1706** (2017) 066, arXiv:1703.04584, Def. 2.1] The **fakeon** (fake particle) is a quantization prescription for a ghost pole at $k^2 = m_f^2$ (negative residue), defined by replacing the Feynman $\mathrm{i}\epsilon$ with the **principal value**:

$$\frac{1}{k^2 - m_f^2 + \mathrm{i}\epsilon} \longrightarrow \mathcal{P}\frac{1}{k^2 - m_f^2} = \frac{1}{2}\left(\frac{1}{k^2 - m_f^2 + \mathrm{i}\epsilon} + \frac{1}{k^2 - m_f^2 - \mathrm{i}\epsilon}\right). \tag{16}$$

Equivalently, the fakeon propagator is the average of the retarded and advanced propagators.

Theorem 5.5 (Perturbative unitarity of fakeon gravity)**.** *[Anselmi & Piva, JHEP **1806** (2018) 039, arXiv:1803.07777; Anselmi, JHEP **2006** (2020) 086, arXiv:2003.12486] The gravitational theory*

$$S = \int \mathrm{d}^4x\,\sqrt{g}\left[\frac{R}{16\pi G} + \alpha\, C_{\mu\nu\rho\sigma}C^{\mu\nu\rho\sigma} + \beta\, R^2\right] \tag{17}$$

*is **perturbatively unitary** when the massive spin-2 mode is quantized with the fakeon prescription* (16)*. Specifically:*

(i) The optical theorem $2\,\mathrm{Im}\,\mathcal{T} = \sum_n |\mathcal{T}_n|^2$ *is satisfied at every loop order, where the sum runs over* physical *(non-fakeon) intermediate states.*

(ii) The theory is renormalizable (Stelle's power-counting argument applies independently of the quantization prescription).

(iii) The fakeon does not appear as an asymptotic state, does not contribute to physical cross-sections, and has zero on-shell residue in the S-matrix.

Remark 5.6 (Applicability to SCT). The SCT ghost pole at $z_0 \approx 2.415$ in Π_{TT} satisfies the prerequisites for the fakeon prescription:

1. The pole is on the *real axis* (required for principal value to be well-defined).
2. The residue is *negative*: $R_2 \approx -0.493$ (the pole is a ghost, not a normal particle).
3. $|R_2| < 1 = |R_2^{\text{Stelle}}|$: the ghost coupling is suppressed relative to the local (Stelle) theory.

[GAP] What remains to be verified:

- Whether the nonlocal structure of $\Pi_{\text{TT}}(z)$ (infinitely many complex zeros from Hadamard factorization) is compatible with the fakeon prescription at higher loops.
- Whether the optical theorem cancellations extend to the nonlocal vertices that arise in SCT.
- The fate of complex conjugate zeros: pairs of complex zeros could correspond to Lee-Wick partner pairs, requiring the CLOP contour prescription rather than the fakeon principal value.

Remark 5.7 (Fakeon vs. CLOP). [Anselmi, arXiv:2003.12486, §4] The Anselmi-Piva fakeon prescription generalizes and replaces the original CLOP prescription. Key differences:

(i) CLOP requires complex conjugate pole pairs; the fakeon applies to real poles as well.
(ii) CLOP gives a specific contour deformation; the fakeon gives a local prescription (principal value) that is easier to implement in loop calculations.
(iii) Both agree for theories where they both apply (complex conjugate poles on the real axis).

For SCT: the real ghost pole at z_0 should use the fakeon prescription; potential complex conjugate zeros (if found) would use the CLOP contour.

6 Wick Rotation in Curved Spacetime

6.1 Visser: Metric Deformation

Convention 6.1 (Visser Wick rotation)**.** [Visser, Phys. Lett. B **159** (1985) 22] For a globally hyperbolic spacetime with ADM decomposition

$$\mathrm{d}s^2 = -N^2\,\mathrm{d}t^2 + h_{ij}(\mathrm{d}x^i + N^i\,\mathrm{d}t)(\mathrm{d}x^j + N^j\,\mathrm{d}t)\,, \tag{18}$$

the **Visser deformation** introduces a parameter $\epsilon \in [-1, +1]$:

$$g_\epsilon = -\epsilon\,N^2\,\mathrm{d}t^2 + h_{ij}(\mathrm{d}x^i + N^i\,\mathrm{d}t)(\mathrm{d}x^j + N^j\,\mathrm{d}t)\,. \tag{19}$$

At $\epsilon = +1$: Lorentzian signature $(-,+,+,+)$ (original metric). At $\epsilon = -1$: Euclidean signature $(+,+,+,+)$.

This is a deformation of the *metric*, not the time coordinate. It avoids the ambiguity of complexifying t in curved spacetime.

Remark 6.2 (Status and limitations). The Visser deformation is a heuristic: there is no proof that the gravitational path integral (or the spectral action) is analytic in ϵ along the full interval $[-1, +1]$. Phase transitions, singularities of the ADM decomposition, or topology changes could produce non-analyticities.

For SCT, the relevant question is: are the form factors $F_i(z)$ analytic under $z \to z(\epsilon)$ as ϵ varies from $+1$ to -1? Since $F_i(z)$ are entire functions of z, the answer is yes *for fixed momentum* k^2. The issue is whether the *spectral action functional* (which integrates over all momenta) is analytic in ϵ.

6.2 Kontsevich-Segal: Allowable Complex Metrics

Definition 6.3 (Allowable metric)**.** [Kontsevich & Segal, arXiv:2105.10161, Def. 1.1] A complex metric g on a smooth manifold M is **allowable** if at every point $x \in M$, the complex symmetric matrix $g_{\mu\nu}(x)$ satisfies:

(i) $\mathrm{Re}\Big(\sqrt{\det g}\Big) > 0$,

(ii) For each eigenvalue λ_i of $g_{\mu\nu}(x)$: $|\arg(\lambda_i)| < \pi/2$.

Theorem 6.4 (Allowable signatures)**.** *[Kontsevich & Segal, arXiv:2105.10161, Thm. 1.3; Witten, arXiv:2111.06514, §3] Among* real *metrics on a d-dimensional manifold, only signatures $(d, 0)$ (Euclidean) and $(d-1, 1)$ (Lorentzian) are allowable. Specifically:*

- *Euclidean metrics (all eigenvalues positive,* $\arg = 0$*): strictly allowable, in the interior of the allowable domain.*
- *Lorentzian metrics (one negative eigenvalue,* $\arg = \pi$*): on the* boundary *of the allowable domain (the negative eigenvalue has* $|\arg| = \pi$*, which saturates the bound but is included as a limiting case).*
- *All other real signatures (e.g.,* $(2, 2)$*,* $(3, 1, 0)$*) are* not *allowable.*

Remark 6.5 (Implications for Wick rotation). The Kontsevich-Segal criterion provides a mathematical justification for the Visser deformation: the one-parameter family g_ϵ with $\epsilon \in [-1, +1]$ passes through only allowable metrics (for $\epsilon > 0$: Euclidean; for $\epsilon = -1$: Lorentzian on the boundary; for $-1 < \epsilon < 0$: complex with $|\arg| < \pi$).

For the gravitational path integral, Kontsevich and Segal conjecture (and Witten elaborates) that restricting to allowable metrics gives a convergent integral. For SCT, this suggests that the effective action computed in Euclidean signature via the spectral action can be analytically continued to Lorentzian signature along the Visser path, *provided* the integrand (form factors) has no singularities along this path.

Since $F_1(z)$ and $F_2(z)$ are entire, they have no singularities anywhere in $\mathbb{C}$, including along the Wick-rotation path. The potential obstruction is the zero of Π_{TT} at $z_0 \approx 2.415$, which produces a pole in the propagator.

7 The Chamseddine-Connes Spectral Action

Definition 7.1 (Spectral action)**.** [Chamseddine & Connes, Commun. Math. Phys. **186** (1997) 731, hep-th/9606001, Eq. (1.1)] For a spectral triple $(\mathcal{A}, \mathcal{H}, D)$ (Riemannian), the **bosonic spectral action** is:

$$S_b = \mathrm{Tr}\Big(f(D^2/\Lambda^2)\Big) + \frac{1}{2}\langle J\Psi, D_A\Psi\rangle\,, \tag{20}$$

where f is a positive even function (cutoff), Λ is the energy scale, $D_A = D + A + JAJ^{-1}$ is the fluctuated Dirac operator, and the second term is the fermionic action.

The bosonic part has the asymptotic expansion for $\Lambda \to \infty$:

$$\mathrm{Tr}\, f(D^2/\Lambda^2) \sim \sum_{k\geq 0} f_k\, \Lambda^{4-k}\, a_k(D^2)\,, \tag{21}$$

where a_k are the Seeley-DeWitt coefficients and $f_k = \int_0^\infty f(u)\, u^{(4-k)/2-1}\, \mathrm{d}u$ are the moments of the cutoff function.

Remark 7.2 (Assumptions requiring Riemannian signature). The spectral action (20) relies on:

(i) D is self-adjoint on a Hilbert space $\Rightarrow D^2 \geq 0$ (non-negative spectrum).
(ii) $f(D^2/\Lambda^2)$ is a bounded operator (since f is a Schwartz-class cutoff and $D^2 \geq 0$).
(iii) The heat kernel e^{-tD^2} is trace-class for $t > 0$, enabling the Seeley-DeWitt expansion.

All three fail in Lorentzian signature:

(i) D is J-self-adjoint on a Krein space, but D^2 has indefinite spectrum.
(ii) $f(D^2/\Lambda^2)$ is not bounded if D^2 has negative eigenvalues and f grows.
(iii) e^{-tD^2} is not trace-class (exponential growth in the negative spectral direction).

Remark 7.3 (Chamseddine-Connes-Marcolli: full Standard Model). [Chamseddine, Connes, Marcolli, hep-th/0610241] The spectral action for the *almost-commutative* geometry $M \times F$ (where F is the finite spectral triple encoding the Standard Model) produces the full Einstein-Hilbert + Standard Model Lagrangian at order Λ^0 in the asymptotic expansion (21). The non-local form factors F_1, F_2 arise from keeping the *exact* heat kernel rather than its asymptotic expansion.

The entire NCG Standard Model programme is formulated in Euclidean signature. Barrett's Lorentzian KO-dimension fix (§2) addresses only the fermionic action; the bosonic spectral action remains Euclidean.

8 Causal Set d'Alembertians

Definition 8.1 (Causal set d'Alembertian). [Sorkin, arXiv:gr-qc/0703099; Benincasa & Dowker, Phys. Rev. Lett. **104** (2010) 181301] On a causal set $\mathcal{C}$ (a locally finite partial order approximating a Lorentzian manifold), the **causal set d'Alembertian** B_ℓ acting on a scalar field is:

$$B_\ell\, \phi(x) = \frac{1}{\ell^2} \sum_{n=0}^{N_d} C_n^{(d)} \sum_{y \prec_n x} \phi(y)\,, \tag{22}$$

where ℓ is the discreteness scale, $\prec_n$ denotes the n-th layer of causal predecessors, and $C_n^{(d)}$ are dimension-dependent coefficients chosen so that $B_\ell \to \Box$ in the continuum limit.

Theorem 8.2 (Nonlocal massive modes in continuum limit). *[Belenchia, Benincasa, Liberati, JHEP **1503** (2015) 036, arXiv:1411.6513, Thm. 1] In the continuum limit of the causal set d'Alembertian B_ℓ, the retarded Green's function acquires a **continuum of massive modes**: the spectral density $\rho(m^2)$ is non-zero for all $m^2 > 0$, even as $\ell \to 0$. These massive modes are* non-perturbative *in ℓ (they survive the continuum limit) and represent a form of discreteness-induced nonlocality.*

The momentum-space propagator of B_ℓ in the continuum limit takes the form:

$$\tilde{G}_\ell(k) = \frac{1}{-k^2 + f_\ell(k^2)}\,, \tag{23}$$

where $f_\ell(k^2)$ is a nonlocal correction that contains the massive spectral density.

Remark 8.3 (Analogy with SCT). The causal set d'Alembertian provides a conceptual parallel to SCT:

- Both produce nonlocal modifications of $\Box$.
- Both generate a continuum of massive modes (in SCT: from the form factors F_1, F_2; in causal sets: from discreteness).

- Both are intrinsically Lorentzian (causal sets have causal order built in; SCT inherits it from the spectral triple).

However, the mathematical frameworks are fundamentally different: causal sets are discrete structures, while SCT is a continuum theory based on spectral geometry. The analogy is suggestive but not directly useful for deriving the Lorentzian SCT action.

9 Conventions Summary

Quantity	**Convention**	**Source**
Krein space	$(\mathcal{K}, [\cdot,\cdot])$, $J^2 = \mathrm{Id}$, $J = J^*$	Bognár 1974
Fundamental symmetry	$J = \mathrm{i}\gamma^0$ (Lorentzian)	Strohmaier
J-self-adjointness	$[D\psi, \phi] = [\psi, D\phi] \Leftrightarrow JDJ = D^*$	Strohmaier
KO-dimension	6 (mod 8) (Lorentzian SM)	Barrett
Lorentzian Dirac	$D = -\mathrm{i}\gamma^\mu \nabla_\mu$	standard
Visser deformation	g_ϵ, $\epsilon \in [-1, +1]$	Visser 1985
Allowable metrics	$\lvert \arg(\lambda_i) \rvert < \pi/2$	Kontsevich-Segal
Ghost-free criterion	$\Pi(z) \neq 0$ for $z \geq 0$	BMS
Fakeon prescription	$\mathcal{P}(k^2 - m_f^2)^{-1}$	Anselmi-Piva
SCT master function	$\phi(z) = \int_0^1 e^{-\xi(1-\xi)z}\, \mathrm{d}\xi$	NT-1
SCT form factors	$\hat{F}_i(z) = F_i(z)/F_i(0)$	NT-4a
Ghost pole	$z_0 \approx 2.415$, $R_2 \approx -0.493$	MR-2 d.5

10 Gap Analysis

G1. Lorentzian spectral action [CRITICAL]: The spectral action $\mathrm{Tr}\, f(D^2/\Lambda^2)$ is undefined in Lorentzian signature because D^2 is hyperbolic with indefinite spectrum. Recent work by Dang and Wrochna [34, 35] defines complex powers of the wave operator and Lorentzian spectral zeta functions on asymptotically Minkowski spacetimes, providing a candidate replacement. However, the connection to the Chamseddine-Connes spectral action (which uses heat kernel regularization, not zeta functions) remains to be established.

Candidate resolutions:

(a) Accept the Euclidean spectral action as fundamental; Wick-rotate the *effective action* (a functional of the metric) via Visser deformation (§6).
(b) Define a distributional or regularized Lorentzian trace (e.g., using the advanced-minus-retarded propagator instead of the heat kernel).
(c) Use the Barvinsky-Vilkovisky in-in formalism, which is intrinsically Lorentzian, to derive the same nonlocal effective action.
(d) Use Lorentzian spectral zeta functions (Dang-Wrochna) as an alternative regularization of the spectral action.

G2. Fate of the ghost pole [CRITICAL]: $\Pi_{\mathrm{TT}}(z_0) = 0$ at $z_0 \approx 2.415$. In Lorentzian signature, this becomes a physical propagating massive spin-2 mode with mass $m_2^2 = z_0 \Lambda^2$ and negative norm.

Candidate resolutions:

(a) Anselmi-Piva fakeon prescription: quantize the ghost as a fakeon (principal value).
(b) Lee-Wick (CLOP) contour: if complex conjugate partner zeros exist, use the CLOP contour for the pair.
(c) Accept the ghost and investigate whether the theory is asymptotically safe (ghost decouples at high energies where $\Pi_{\mathrm{TT}} \to 1$).

G3. Complex zero structure of Π_{TT}: $\Pi_{\mathrm{TT}}(z)$ is entire of order 1. By Hadamard's theorem it has infinitely many zeros. The complex zeros (off the real axis) will become relevant in Lorentzian continuation: pairs of complex conjugate zeros correspond to Lee-Wick partner pairs.

Required: Systematic numerical survey of zeros in the complex z-plane, classification into real/complex conjugate pairs, and determination of which prescription (fakeon or CLOP) applies to each.

G4. Wick rotation analyticity: The Visser deformation g_ϵ provides a path from Euclidean to Lorentzian, but there is no proof that the spectral action (or effective action) is analytic in ϵ along the entire path. Non-analyticities could arise from: phase transitions in the path integral, singularities of the ADM decomposition, or level crossings in the Dirac spectrum.

G5. Scalar mode at $\xi \neq 1/6$: At conformal coupling $\xi = 1/6$, $\Pi_s = 1$ (no scalar ghost). For $\xi \neq 1/6$, Π_s develops its own zero structure, producing additional ghost poles. The full Lorentzian theory must address both spin-2 and spin-0 ghosts.

G6. Non-perturbative definition: All verified SCT results are perturbative (one-loop, $O(\mathcal{R}^2)$). A complete Lorentzian formulation would require either a non-perturbative spectral action or a proof that perturbative results capture the essential physics.

G7. Higher-loop stability: The fakeon prescription is proved for perturbative unitarity at *all* loop orders for the local Stelle action. For the nonlocal SCT action, the proof has not been extended beyond tree level.

11 Comparison of Approaches

Approach	Spectral action	Ghost handling	SCT fit	Maturity
Krein space (Strohmaier, van den Dungen)	Fermionic: defined. Bosonic: **undefined**	Not addressed	Correct framework for D	Medium
Wick rotation (Visser, KS)	Defined in Euclidean; continued via g_ϵ	Poles rotate in complex k^2 plane	Natural bridge for existing results	Low–Medium
Fakeon (Anselmi-Piva)	N/A (prescription for propagator)	Ghost $\to$ fakeon: removed from spectrum	Directly addresses Π_{TT} ghost	High

Approach	Spectral action	Ghost handling	SCT fit	Maturity
CLOP / Lee-Wick	N/A	Complex conjugate pairs: contour deformation	Applies if complex zeros pair up	High
Ghost-free IDG (BMS, Tomboulis)	N/A (action design)	$a(z) = e^{H(z)}$: no zeros by construction	SCT does NOT satisfy	High
Causal sets (Sorkin)	Different framework	Nonlocality = UV regulator	Conceptual analogy only	Low

12 Recommended Strategy for MR-1 Derivation

Based on the literature survey, we recommend a three-component approach to the Lorentzian continuation of SCT:

(A) Krein-space foundation: Adopt Strohmaier's pseudo-Riemannian spectral triple (Definition 2.4) as the mathematical framework. Barrett's KO-dimension fix (Theorem 2.8) resolves fermion doubling. The fermionic action is well-defined in Lorentzian signature.

(B) Effective action via Wick rotation: For the bosonic sector, accept the Euclidean spectral action as the fundamental definition. Wick-rotate the *effective action* (functional of $g_{\mu\nu}$) via Visser deformation (Convention 6.1). The form factors are entire functions, hence well-defined under analytic continuation. Verify that the Kontsevich-Segal criterion (Theorem 6.4) is satisfied along the deformation path.

(C) Fakeon prescription for the ghost pole: Quantize the spin-2 ghost at $z_0 \approx 2.415$ using the Anselmi-Piva fakeon prescription (Definition 5.4). The prerequisites are satisfied: real pole, negative residue, suppressed coupling. For complex zeros (if found in pairs), use the CLOP contour.

The deliverable of MR-1 is a self-consistent Lorentzian formulation of SCT at the one-loop, $O(\mathcal{R}^2)$ level, with:

- A well-defined kinematic framework (Krein spectral triple),
- A well-defined bosonic action (Wick-rotated effective action),
- A ghost-free physical spectrum (fakeon prescription),
- Agreement with all verified Euclidean results in the appropriate limit.

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